

# Hillclimbing

What it does, step by step — applied to dotspeak

Hillclimbing is a loop that makes small changes and keeps only the ones that measurably help. It needs two things: a **score**, and a **knob** the loop can turn and re-measure (usually a prompt Claude uses). The example below is a “sentence smoother” that turns picked words like *I go home hour* into a fluent sentence.

## Setup (done once)

- 1 Collect test cases.** For example 40 word sequences someone might pick in dotspeak, such as *I go home hour* or *man hate cold*. You review them and confirm they are realistic.
- 2 Decide how to grade.** Each output gets a score. Here a second Claude call (the “judge”) checks a short list of rules: all words used, meaning kept, sounds natural. You look at a few graded examples and confirm you would have scored them the same way.
- 3 Split the cases.** Most go in a *train* set the loop learns from. A few are held back as a *test* set it never looks at, to catch changes that only fit the train cases.
- 4 Baseline run.** Run the current prompt on all cases and record the score, for example 62%.

## The loop (repeats)

- 5 Read the failures.** Look at the cases that scored badly, e.g. “it keeps dropping the word *hour*”.
- 6 Propose one change.** For example, add to the prompt: “Use every word exactly once, in order.”
- 7 Re-run all cases** with the changed prompt.
- 8 Compare.** If the score went up by more than the noise (e.g. 62% → 74%), keep the change; it becomes the new baseline. If it went down or stayed flat, undo it.
- 9 Record it.** Each attempt is saved as v1, v2, ... with what changed and its score, so the whole path shows up in a report.
- 10 Go back to step 5** until the score stops improving, or the budget or round limit you set is reached.

## Finish

- 11 Check the test set.** If the held-back cases also improved, the gains are real. If only the train cases improved, the prompt was tuned to those specific examples.

## Cost

Every run calls Claude once per case, plus once more for the judge. With 40 short cases, one round costs a few cents. You approve the budget before any paid run.

## Possible knobs for dotspeak

| Option                     | What the loop improves                                                               | How it is scored                                                                                                |
|----------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Sentence smoother</b>   | The prompt of a new Claude feature that turns picked words into a fluent sentence.   | Claude judge with a rubric: all words used, meaning kept, natural.                                              |
| <b>Word lists</b>          | The .txt files themselves; the loop proposes edits.                                  | Claude judge scores category fit, coverage and duplicates. Risk: it optimizes for the judge’s taste, not yours. |
| <b>Word-to-code finder</b> | The prompt of a new Claude feature that predicts which code (e.g. 7 7) holds a word. | Exact match against your existing files: free and objective.                                                    |

Note: dotspeak does not use Claude yet, so the first step for any option is adding the Claude feature itself.

## Why isn't it used everywhere?

I agree it's underused. The loop is simple and works on almost anything: prompts, configs, word lists, even non-AI settings like the font size in your app. The idea is decades old, from optimization. What's new is that Claude can do the slow parts itself: read the failures, propose a change, and run the eval.

"Everywhere" comes with one condition, though: you need a score you trust. Hillclimbing pushes hard toward whatever the score rewards. If the judge's rubric is slightly off, the loop will find that gap and exploit it, and the number rises while real quality doesn't. That's why the setup steps (reviewing the test cases yourself, checking the judge's grades, keeping a held-back test set) matter more than the loop. Building that eval is the hard part, and it's probably what makes hillclimbing harder to promote than it sounds.